

Table YE-1 Definitions for the Different Interface Points

	<i>POS</i> <i>Delivery Point of Gas/Chemical Supplier</i>	<i>POD</i> <i>Outlet of Central Facility System</i>	<i>POC</i> <i>Submain or VMB/VMP Take off Valve</i>	<i>POE</i> <i>Entry to Equipment or Sub Equipment</i>	<i>POU</i> <i>Entry to the Process Chamber</i>	<i>POP</i> <i>Contact with Wafer</i>
Ultrapure water	Raw water	Outlet of final filtration in UPW plant	Outlet of submain take off valve	Inlet of wet bench or sub-equipment (downstream POU filter, if available)	Inlet of wet bench bath, spray nozzle, or connection point to piping, which is also used for other chemicals	Wafer in production
Process chemicals	Chemical drum/tote/bulk supply	Outlet of final filtration of chemical distribution unit	Outlet of VMB valve	Inlet of wet bench or intermediate tank	Inlet of wet bench bath or spray nozzle	Wafer in production
Specialty gases	Gas cylinder or bulk specialty gas systems	Outlet of final filtration of gas cabinet	Outlet of VMB valve	Inlet of equipment	Inlet of chamber (outlet of MFC)	Wafer in production
Bulk gases	Bulk gas delivered on site or gas generator	Outlet of final filtration/purification	Outlet of submain take off valve or VMB valve	Inlet of equipment/sub-equipment	Inlet of chamber (outlet of MFC)	Wafer in production
Cleanroom and AMC	Outside air	Outlet of make-up air handling unit	Outlet of filters in cleanroom ceiling	Inlet to mini-environment or sub equipment for AMC, outlet of the tool filter for particles	Gas/air in vicinity to wafer/substrate	Wafer/substrate in production (AMC/SMC)

POD—point of delivery POC—point of connection POE—point of entry POP—point of process POU—point of use
VMB—valve manifold box VMP—valve manifold post UPW—ultra pure water MFC—mass flow controller
AMC—airborne molecular contamination SMC—surface molecular contamination

Characterization, Inspection and Analysis—Physical device dimensions and corresponding defect dimensions continue shrinking, posing new challenges to detection as well as tolerable contamination. The wafer edges and backside were identified to show significant impact on yield as well as process variations and design. Development of defect detection, defect review, and classification technologies showing highest sensitivity at high throughput is crucial for cost efficient manufacturing. Furthermore, for efficient manufacturing the monitoring of contamination in the environment and on the wafer, surface requires appropriate analytic capabilities. Automated, intelligent analysis and reduction algorithms, which correlate facility, design, process, electrical and virtual metrology results, and their correlation to yield, test and work-in-progress data, will have to be developed to enhance root cause analysis and therefore enable rapid yield learning.

3. SUMMARY AND KEY POINTS

It is recommended that the reader of the IRDS Yield Enhancement Chapter reviews this narrative along with other materials and particularly Table YE-3, including associated notes. It is important to keep in mind that although the values in the roadmap were defined based on empirical data and modeling, there is no sufficient ground to justify specific technological decision, which are expected to be made by the technology end users based on the process specific sensitivities and requirements.

It is also important to note that definition of the semiconductor technology needs in the roadmap does not take into account feasibility of existing facility technology and metrology capability. Applying the roadmap values for facility systems specifications may pose execution challenges. Instead, it is recommended to refer to definitions of most current SEMI standards (if available) where the feasibility issues are taken into account and solutions are recommended, under consideration of the roadmap definitions.

Given limitations of the metrology used in material, environment, and process control, it is recommended to focus on proactive measures related to technology management. Such approach should lean on standardization, data mining, and process performance optimizations related to all aspects of the facility, materials, and other technologies involved in the Yield enabling and enhancement. Particles control in liquid chemicals and UPW is an example of such challenge.

4. CHALLENGES

4.1. NEAR-TERM CHALLENGES —CIA

Currently, one of the important key challenges will be the detection of multiple killer defects and the signal-to-noise ratio. It is a challenge to detect multiple killer defects and to differentiate them simultaneously at high capture rates, low cost of ownership and high throughput. Furthermore, it is difficult to identify yield relevant defects under a vast amount of nuisance and false defects. As another challenge with high priority the requirement for 3D inspection was identified. This necessitates for inspection tools with the capability to inspect high aspect ratios but also to detect non-visuals such as voids, embedded defects, and sub-surface defects is crucial. The demand for high-speed and cost-effective inspection tools remains, especially in the area of 3D inspection as the importance of 3D defect types increases. In subchapter to Characterization, Inspection and Analysis another key challenge was identified: detection of organic contamination on surfaces – the detection and characterization of non-volatile organics on surfaces is currently not possible in the fab. There are few laboratories or fab scale instrumentation available or implemented except ToF-SIMS, and XPS.

MEMS technologies have different challenges. Inspection and review tools need to be able to handle a broad range of wafer thicknesses. Infrared inspection is required as an automatic and full wafer scan option, which is available in principle but has not yet proven to meet the required throughput requirements. Review options need to be able to have solely edge grip handling and an automatic focus adjustment for a considerable wafer bow.

Traditional yield management focuses on adequate inline inspection capabilities to detect and control all relevant defect types to set up short feedback loops as well as enable correlation to yield and test fails. To move to a more proactive approach, a complete range of inline-produced data needs to be used, with the ultimate goal to get signals before hardware is affected. Therefore, a system for yield prediction to achieve a proactive yield perfection based on a full fab data transparency is needed. The data used are MES/wafer tracking data, trace (streaming data from machines), metrology, defectivity, PCM data, yield data (wafer probe, binsort, final test), consumables/raw material data, facility data, environment data, material release data, device and product data, failure analysis data, reliability data, and equipment/maintenance (log files).

The goal will be the improvement of process stability, yield and reliability through a holistic approach for data analysis. Major challenges are the connection of all relevant data sources, Continuous learning from unknowns, how to incorporate subject matter expertise (SME), the definition of a way to structure and facilitate the capture of relevant information (e.g., FDC trace data) to bring value for analysis, traceability, ownership and responsibility on data (overcome ownership siloing)

Other topics challenging the Yield Enhancement community are listed as follows for the focus in the near term:

- Process stability versus absolute contamination level
- Wafer edge, backside and bevel monitoring and contamination control
- Development of sub 10 nm water and chemical liquid particle counter as well as tight particle control technologies (new, tighter filters with low particles shedding and minimal contribution of metal and organic species)
- Correlation yield and contamination levels

Data, test structures, and methods are needed for correlating critical tool parts quality as well as process fluid contamination types and levels to yield and to determine the required control limits. The issues for this challenge are to define the relative importance of different contaminants to wafer yield, a standard test for yield/parametric effect, and a maximum process variation (control limits). The fundamental challenge is to understand the correlation between impurity concentration in key process steps and device yield, reliability, and performance. This correlation will determine whether further increases in contamination limits are truly required. The challenge increases in complexity as the range of process materials widens and selection of the most sensitive processes for study will be required for meaningful progress.

Furthermore, in the long term the following key challenges were identified:

- Inline defect characterization and analysis

The difficult challenges for the Yield Enhancement chapter are summarized in Table YE-2. Currently, the most important key challenge will be the detection of multiple killer defects and the signal-to-noise ratio. It is a challenge to detect multiple killer defects and to differentiate them simultaneously at high capture rates, low cost of ownership and high throughput. Furthermore, it is difficult to identify yield relevant defects under a vast amount of nuisance and false defects. As a challenge with second priority the requirement for 3D inspection was identified. This necessitates for inspection tools the capability to inspect high aspect ratios but also to detect non-visuals such as voids, embedded defects, and sub-surface defects is crucial. The demand for high-speed and cost-effective inspection tools remains, especially in the area of 3D inspection as the importance of 3D defect types increases. In 2011 and with the change of the scope of the subchapter to Characterization, Inspection and Analysis a new key challenge was identified: Detection of organic contamination on surfaces – the detection and speciation of non-volatile organics on surfaces is currently not possible in the fab. There is no laboratory or fab scale instrumentation available or implemented. In 2020 IRDS studied organic precursor deposition on the wafer surface. These studies were done to determine the risks associated with organic contamination on the wafer surface. Results were published in 2021.

Other topics challenging the Yield Enhancement community are prioritized as follows in the near term:

- Process Stability versus Absolute Contamination Level
- Wafer Edge, Backside and Bevel Monitoring and Contamination Control
- Development of sub 10 nm water and chemical liquid particle counter
- Correlation Yield and Contamination Levels

Furthermore, in the long term the following key challenges were identified:

- Inline defect characterization and analysis
- Next generation lithography

Table YE-2 Yield Enhancement Difficult Challenges

<i>Near-Term Challenges: 2024 2027</i>	<i>Description</i>
Challenge #1 Control of Electrically Active and other particles in Ultrapure Water and Liquid Chemicals for advanced semiconductor manufacturing (logic key driver)	Insufficient metrology and marginal treatment technology capability
Challenge #2 Control of Electrically Active and other particles in UPW of EUV mask production	Insufficient metrology and marginal treatment technology capability
Challenge #3 Critical Organics in UPW	Insufficient metrology for online speciation of organics
Challenge #4 Metals in H ₂ O ₂	Insufficient metrology capability. Characterization needed for specific critical metals for key process steps and segments.
Challenge #5 Slurry particle characterization and other impurity monitoring	Insufficient metrology capability
Challenge #6 Plating chemicals: particles > critical size (>0.150 µm) (#/ml)	Need reliable monitoring of large particles
Challenge #7 Liquid Chemicals: particles > critical size (#/ml)	Need reliable monitoring of particles <10 nm. Need Filtration technology that is reliable at the stated retention rating and that has minimal "rise up" contributions at the time of install into the process.

<i>Near-Term Challenges: 2024–2027</i>	<i>Description</i>
Challenge #8 CMP Slurries: large particles > MPS (#/ml)	Need reliable LPC metrology that can measure targeted particles sizes and concentrations that ignore the MPS sizes and do not require sample dilution to obtain the measurement.
Challenge #9 Post-CMP (pCMP) Cleans particles	Some pCMP cleans will have volatile or highly reactive ingredients that form bubbles in solution, therefore; new metrology and new techniques will be needed to differentiate between bubbles and real particles for contamination control. The pCMP cleans that contain surfactants will have micelle particles and structures that are necessary for the chemistry to perform the intended function. The need is for reliable and accurate metrology to distinguish micelles from other foreign particles in solution.
Challenge #10 Particle measurement for ISO class 1	Need more efficient and effective laser particle counters with detection limit $\ll 100$ nm or CPC with sample flow $\gg 0,1$ cfm
Challenge #11 AMC metrology	Need of more sensitive gas analyzers for HCl, HF, HNO ₃ , HBr, NH ₃ , Total Acids, H ₂ S, SO ₂ , organic acids, sulfuric acids and Total bases with detection limit $\ll 100$ ppt
Challenge #12 Particle contribution by critical components and its control	Insufficient analytical capability and unknown level of cleanliness
Challenge #13 Critical metals in UPW for CMOS image sensors	Insufficient metrology sensitivity below 200 ppq concentration
Challenge #14 Low concentration particle measurement in the 10-100nm range to check the cleanliness of gases and to judge the contribution to particle cleanliness from hardware present in the gas delivery system (upstream or downstream from POD).	Preferably these measurements should be possible at a freely selectable flowrate instead of fixed 1 cfm or 0.1 cfm
Challenge #15 Composition and/or structural information for <100nm particles, especially when present in small numbers.	Low amounts of particles have the potential of creating issues. Having excess to techniques that can learn about elemental composition and/or chemical structures can help a lot in root cause finding of observed issues.
Challenge #16 Particulate precursor metrology < 10 nm	Need better understanding of particle precursors and critical concentrations to form particles. Important to determine relationship between species/concentrations and particle size.
Challenge #17 Removing particles < 30 nm	Overcoming the static/van der Waals forces that make particles adhere to critical surfaces is much more difficult for particles < 30 nm.
<i>Long-Term Challenges: 2029–2037</i>	<i>Description</i>
OUT OF CURRENT SCOPE OF YIELD ENHANCEMENT	